Labour Demand 1: Competitive Labour Markets

EC306- Wages and Employment

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Goals of This Section

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- Understand how firms decide how much labour to employ to produce
- Compare short-run vs. long-run labour demand decisions
- Explain why labour demand is downward sloping
- Learn factors that affect the elasticity of labour demand
- Understand the impact of free trade and international competition on the Canadian labour market

Background

Background

- Labour demand refers to the number of workers that firms want to hire at a given wage rate
- It is a **derived demand**
  - Linked to demand for the goods/services produced
  - It is an input into that production
- Several factors will influence labour demand
  - Firm's objectives and constraints
  - Demand conditions in product markets
  - State of technology
  - Time horizon
    - **Short run:** one or more inputs fixed (usually capital)
    - **Long run:** all inputs variable

Structure of Product Markets

- We noted that labour demand is driven by the firm's output
  - Labour is used as a input
- The structure of the product market will influence the demand for labour
  - Ranges from perfect competition (many sellers) to monopoly (one seller)
- By contrast, the structure of the labour market will influence the supply the firm faces
  - Ranges from perfect competition (many workers) to monopsony (one buyer)
- In this section we examine the case of perfect competition in both product and labour markets

Demand in the Short Run

Model

- As labour demand is derived from production, the model starts with a production function
  - Relates inputs to outputs

$$Q = f(K, N)$$

- Where
  - Q is quantity of output
  - K is capital (fixed in short run at K_0)
  - N is labour (variable in short run)
- Firms objective is to maximize profit
- In the short run this means choosing N given fixed K_0
- Optimality is where the marginal revenue product of labour equals its price

Model

- To see this mathematically, take output prices are taken as given (perfect competition).
- Profit

$$\pi(N) = pQ - wN - rK_0$$

$$= pF(K, N) - wN - rK_0$$

- Differentiate $\pi(N)$ w.r.t. N:

$$\frac{d\pi}{dN} = p\frac{\partial F}{\partial N}(K_0, N) - w$$

$$= pMP_N - w$$

Model

- Optimal N^* satisfies

$$pMP_N = w$$

- Where
  - MP_N is the marginal product of labour
  - pMP_N is the marginal revenue product of labour (MRP)
    - When firm is in perfectly competitive product market and $MR = p$ it is called value of marginal product (VMP)
  - w is the wage rate (marginal cost of labour, MC_N)
- Optimality requires that the value of the last unit of labour hired equals its cost

Model

- Other key metrics include
  - Total Revenue Product(TRP): Total revenue associated with labour employed

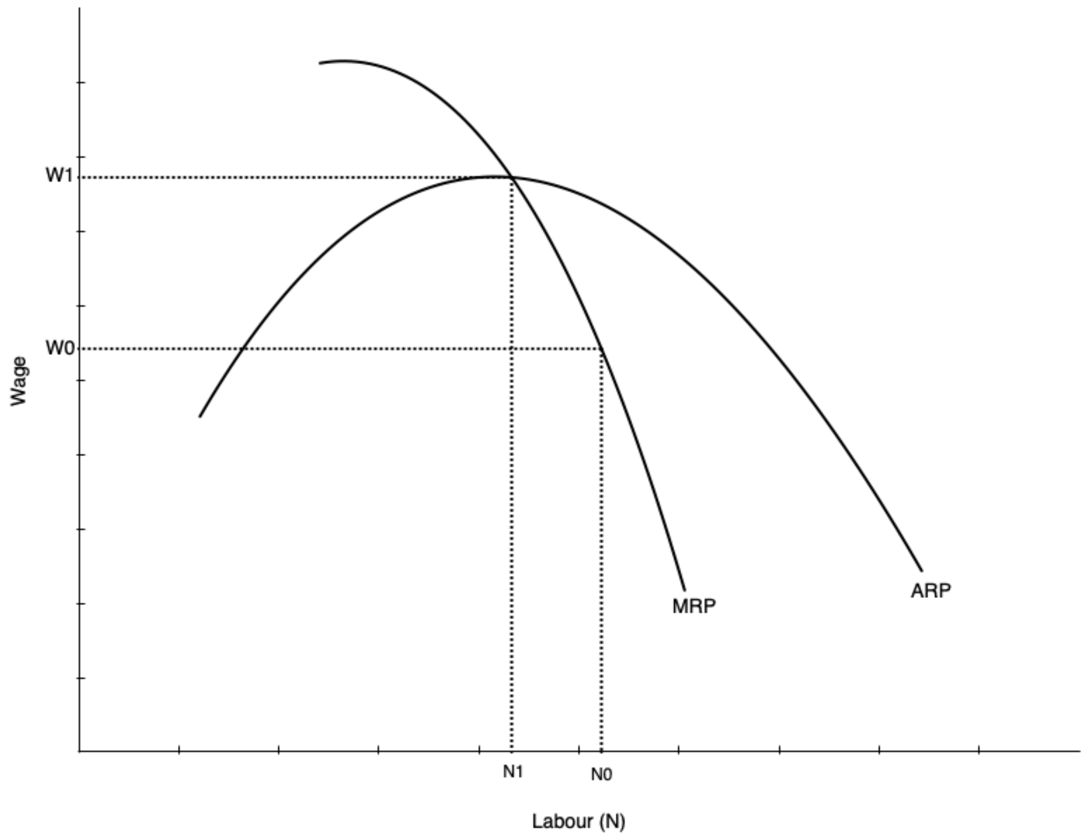
$$TRP = p \times Q$$

  - Average Revenue Product(ARP): Average revenue per unit of labour employed

$$ARP = \frac{TRP}{N} = \frac{p \times Q}{N}$$

- As noted, the firm's optimal choice of labour is where $MRP_N = w$
  - This defines its labour demand curve in the short run
  - As w varies, the firm will adjust N to maintain equality

Labour Demand Curve in Short Run



- The short run labour demand curve is the MRP curve below ARP
  - Says the amount of labour demanded at every wage
- Slopes down because of diminishing marginal product of labour
  - As more labour is employed, extra amount produced falls
  - Not because of inferior labour, but because more labour is using fixed capital
- As wage falls, firm is willing to use more labour

Demand Curve and Competition in Product Market

- The demand for labour depends on the demand for the product being produced
- It therefore also depends on the structure of the product market
- Firm will always maximize profit by choosing labour where $MRP_N = w$
- But marginal revenue in perfectly competitive market and monopolistic market differ
  - In perfect competition $MR = p$, but in monopoly $MR < p$
  - In perfect competition MR does not fall with more N, but in monopoly MR falls as N increases
- So with less competition in product market
  - Labour demand is lower
  - It falls faster with more N because MR falls as N increases

Demand in the Long Run

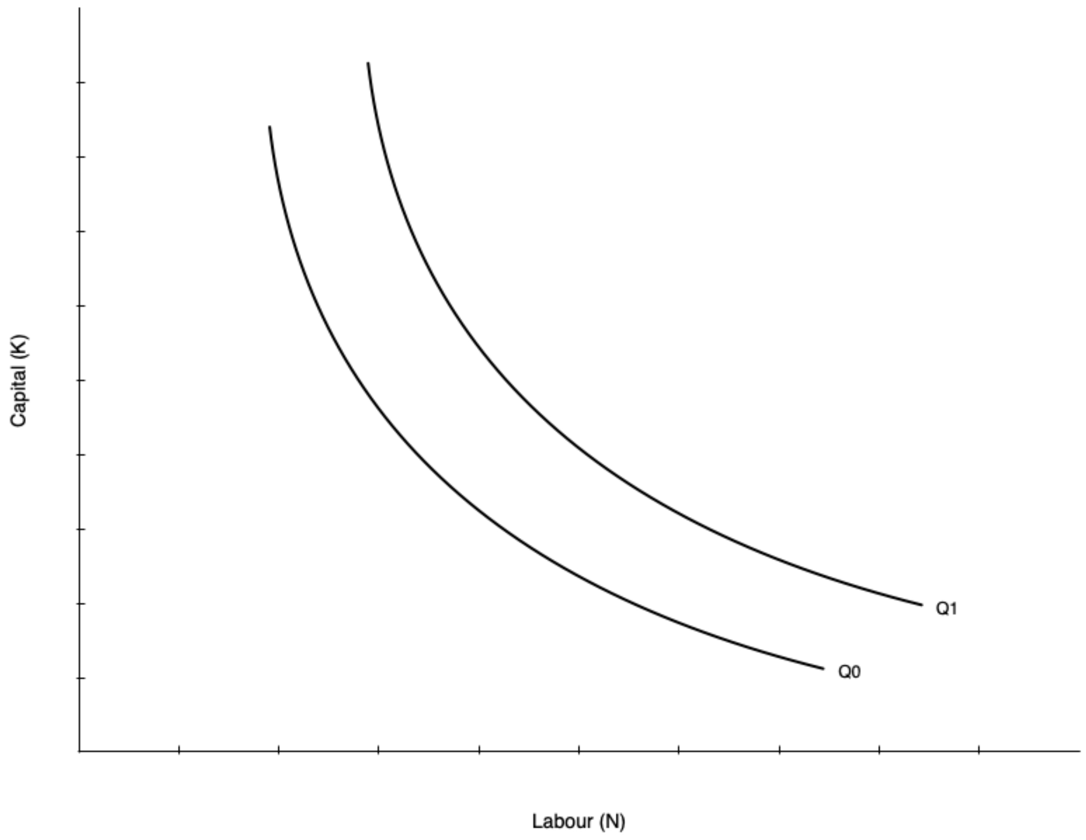
Introduction

- The long run differs from the short run because all inputs are variable
  - Firms can adjust capital as well as labour
- This makes the decision more complicated
  - We now have to consider subsitution between labour and capital
  - We also have to consider economies of scale
- Approach to find optimum occurs in two steps
  - Minimize cost holding output constant
  - Choose output to maximize profit given cost function

Cost Minimization

- First step is to minimize cost of producing given output Q_0
- This occurs when an isoquant is tangent to an isocost line
  - **Isoquant** shows combinations of K and N that produce Q_0
  - **Isocost** line shows combinations of K and N that cost the same amount
  - Analogous to utility maximization with indifference curves and budget constraints
- Assume the cost of capital is r and cost of labour is w
  - Total cost is $C = rK + wN$
- The prouction function is $Q = F(K, N)$
  - We want to minimize C subject to $Q = Q_0$

Cost Minimization



- Graph on left shows two isoquants
  - Combinations of capital and labour that produce the same quantity
- Shape of isoquant is given by the production function
  - Shown as convex, but could be different shapes
  - A linear isoquant would imply perfect substitutes
  - An L-shaped isoquant would imply perfect complements
- Slope of isoquant is the marginal rate of technical substitution (MRTS)
  - MRTS is the rate at which you can

Cost Minimization

- The slope of the isoquant is given by the MRTS

$$MRTS = \frac{MP_N}{MP_K}$$

- Tells you amount of capital you can give up for one more unit of labour holding output constant
- Isoquants are typically display diminishing MRTS
  - When a lot of capital is used, you can give up a lot of capital for one more unit of labour
  - When a lot of labour is used, you can give up only a little capital for one more unit of labour

Cost Minimization

- The other part of cost minimization is the isocost line
  - Gives combinations of K and N that cost the same amount
- Total cost is given by

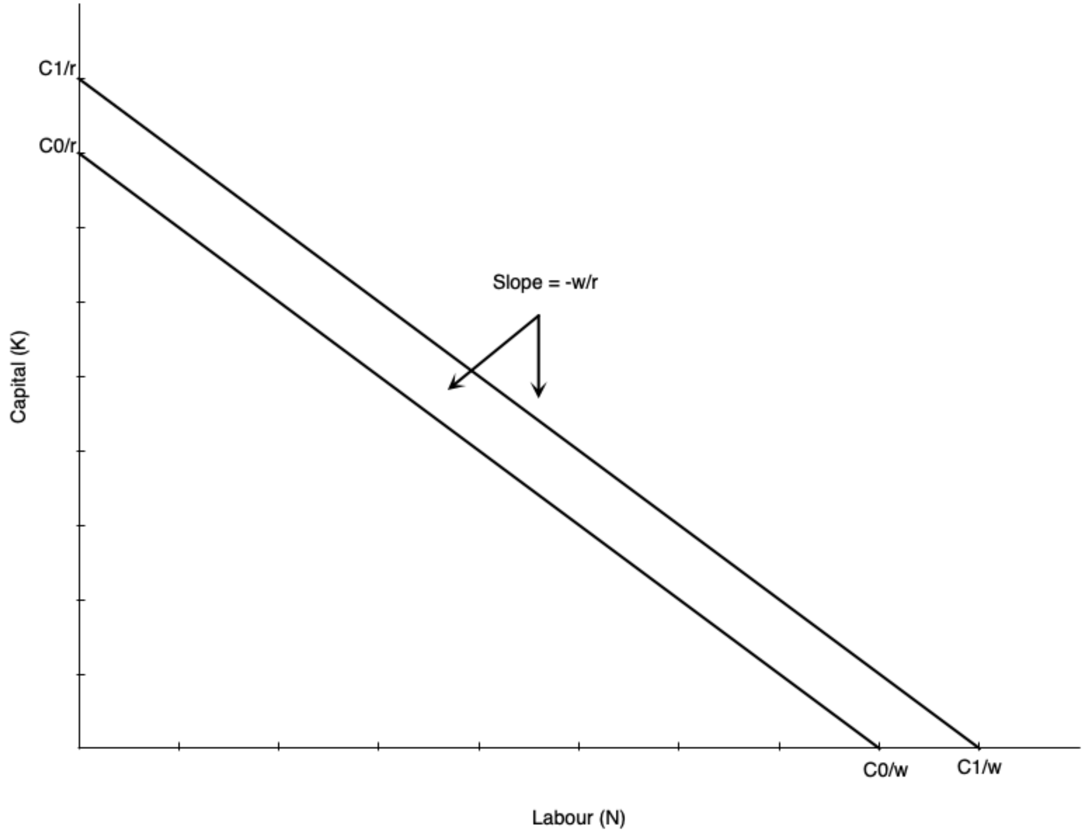
$$C = rK + wN$$

- Rearranging gives the isocost line equation:

$$K = \frac{C}{r} - \frac{w}{r}N$$

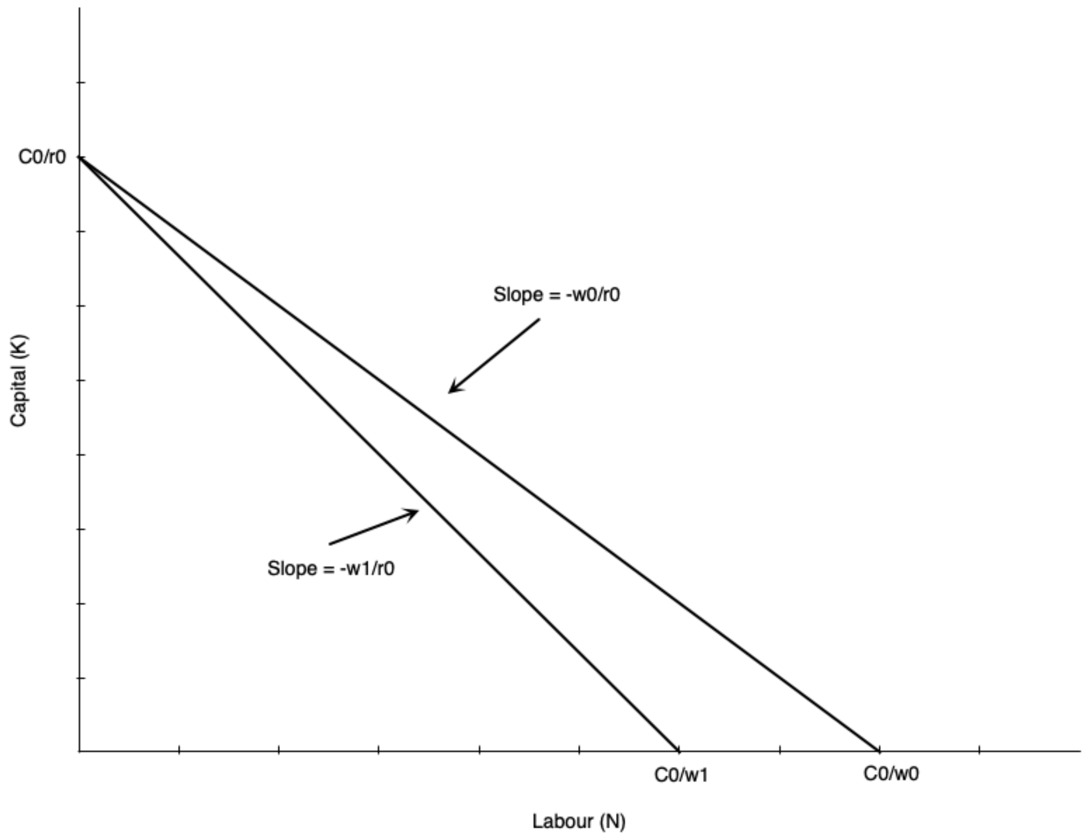
- Slope of isocost line is given by the ratio of input prices
  - As w increases, isocost line becomes steeper
  - As r increases, isocost line becomes flatter

Cost Minimization



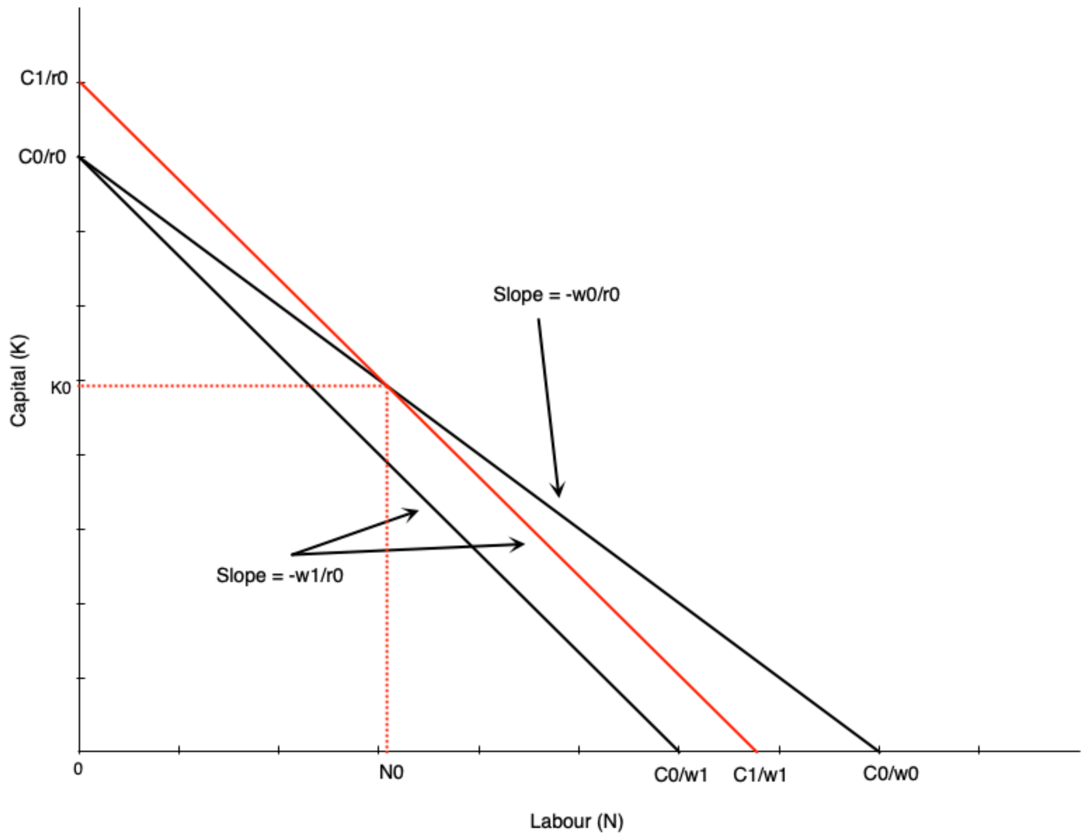
- Graph shows an example isocost
  - Slope is given by ratio of input prices $-w/r$
- If a firm uses only labour, it is equal to C/w
- If a firm uses only capital, it is equal to C/r
- As labour falls by one unit, capital must rise by w/r units to keep cost constant
- Higher cost line shifts isocost line outward (e.g. when C increases to C')

Cost Minimization



- Suppose the wage rises from w_0 to w_1
- Imagine first that we hold cost constant at C_0
- Isocost line would pivot inward
  - Steeper slope as w/r increases
  - If firm uses only capital, vertical intercept remains the same at C_0/r
  - If firm uses only labour, horizontal intercept falls to C_0/w_1
- All combinations of new isocost line lie below old isocost line

Cost Minimization



- To use any of the old combinations of K and N, the firm must increase cost
  - Shifts isocost outward to C_1
  - Magnitude of shift depends on which combination is chosen
- One example depicted to the left
- To use combination (N_0, K_0) at new prices the cost curve shifts out

Cost Minimization

- The cost minimizing bundle is where the isoquant is tangent to the isocost line
- In mathematical terms this means

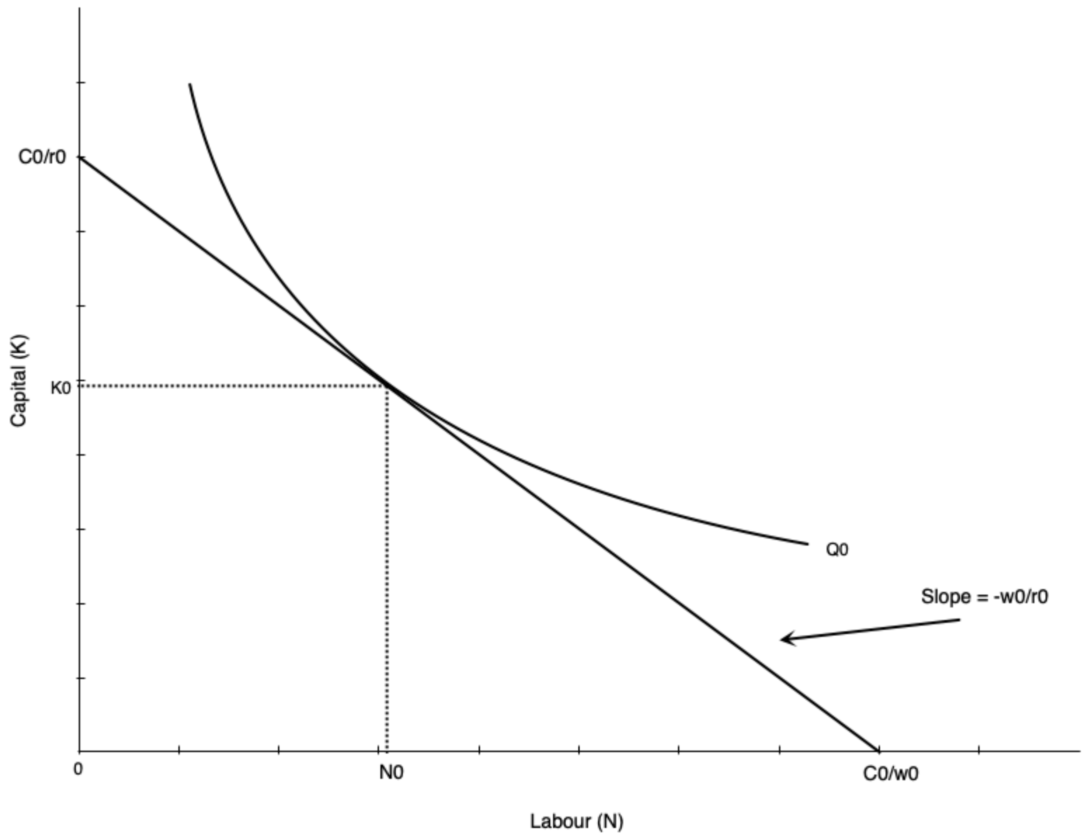
$$\frac{MP_N}{MP_K} = \frac{w}{r}$$

- Says that MRTS is equal to the ratio of input prices
- Alternatively, says that

$$\frac{MP_N}{w} = \frac{MP_K}{r}$$

- The last dollar spent on each input must yield the same marginal product

Cost Minimization



- Cost minimization depicted to the left
- If we hold output constant at Q_0, find the optimal combination of K and N
- Happens at the tangency point between isoquant and isocost line

Cost Minimization

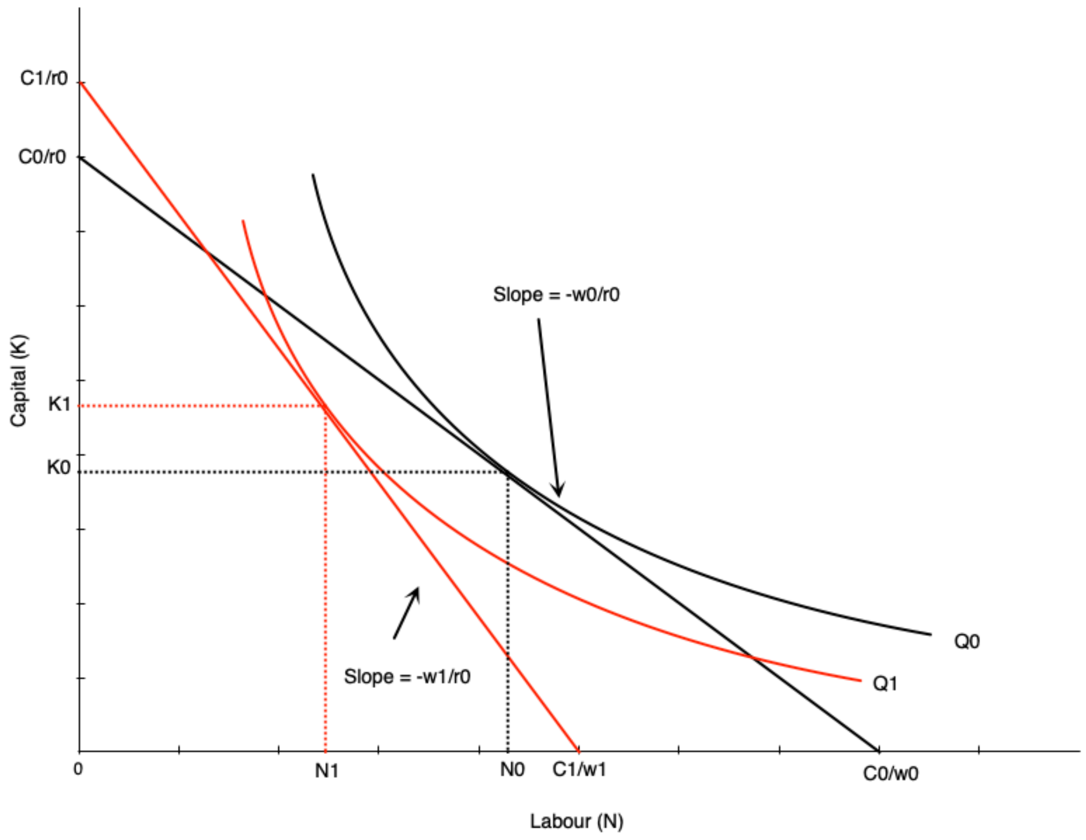
- Note that we have only done cost minimization here
  - Lowest cost given output level Q_0
- Profit maximization involves choosing output level as well
  - Choose Q to maximize profit given cost function
- The second step of choosing output is not depicted here
- If the wage rate goes up, the firm will re-optimize
  - Firm marginal and average cost curves shift up
  - Each firm reduces output
  - Market price rises
- The isoquants at the profit max will shift inward

Deriving Labour Demand in Long Run

Deriving Labour Demand

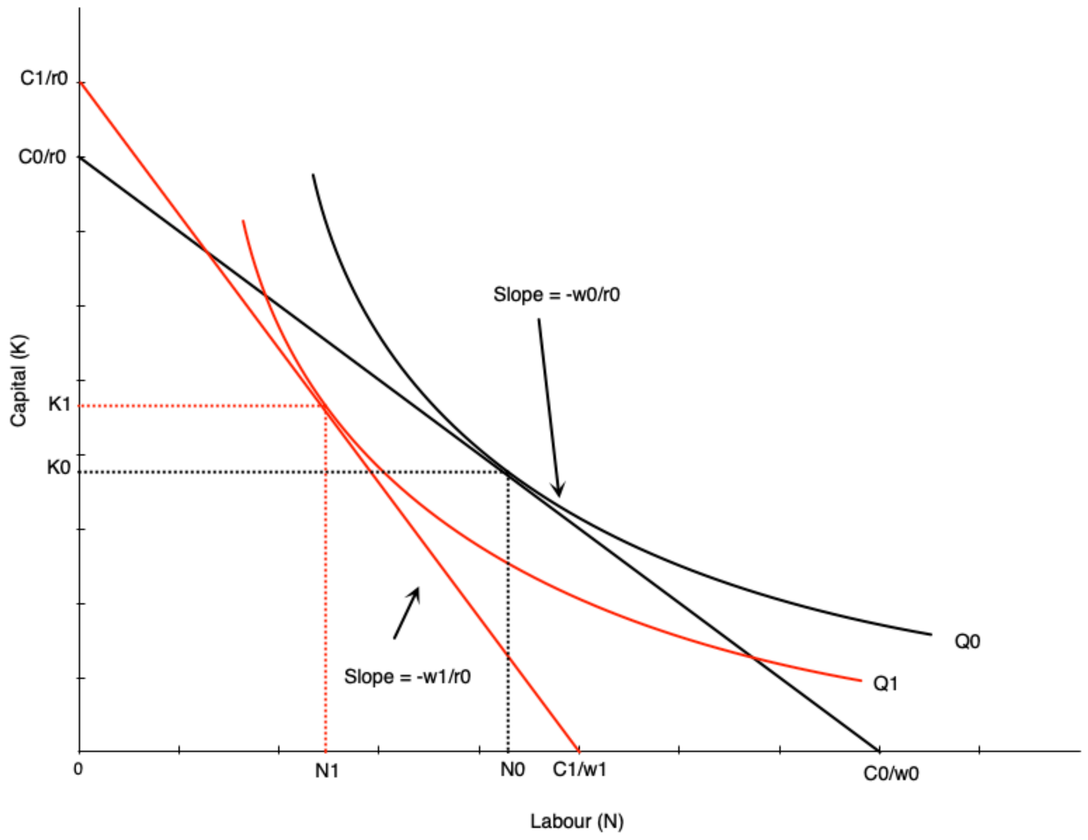
- The labour demand curve plots combinations of w and N such that the firm is maximizing profit
- To derive it, we consider changing wage from w_0 to w_1
  - Isocosts get steeper as w/r increases
  - Firm finds cost minimizing combination of K and N at new wage for each level of output
  - Because cost increases, firm reduces output (optimal isoquant is to the left of original one)
- At new optimum we have
  - Lower quantity of labour demanded
  - Higher wage

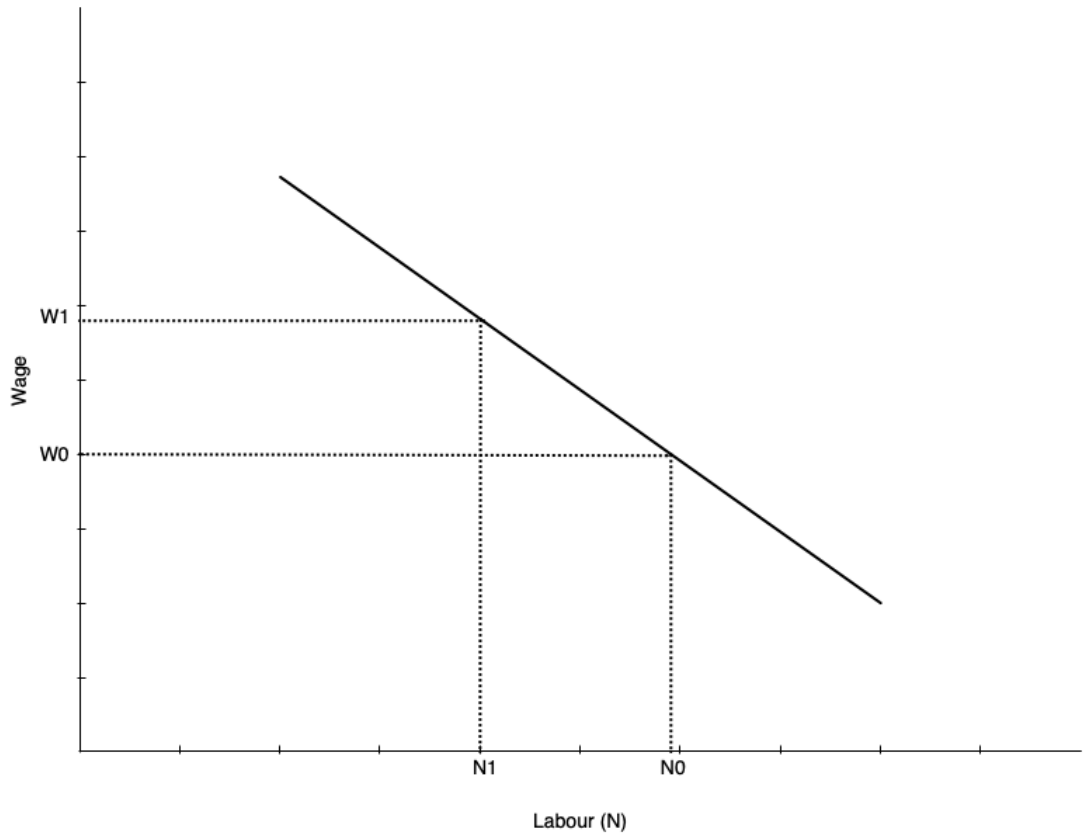
Deriving Labour Demand



- Effect of wage increase on labour depicted to the left
- Wage goes up, which steepens isocost line
- Firm scales down production to lower isoquant
- The tangency between new isoquant and new isocost line shows new optimal labour and capital
- This involves less labour employed
- Effect on capital depends on substitutability between labour and capital
  - The shape of the isoquant

Deriving Labour Demand





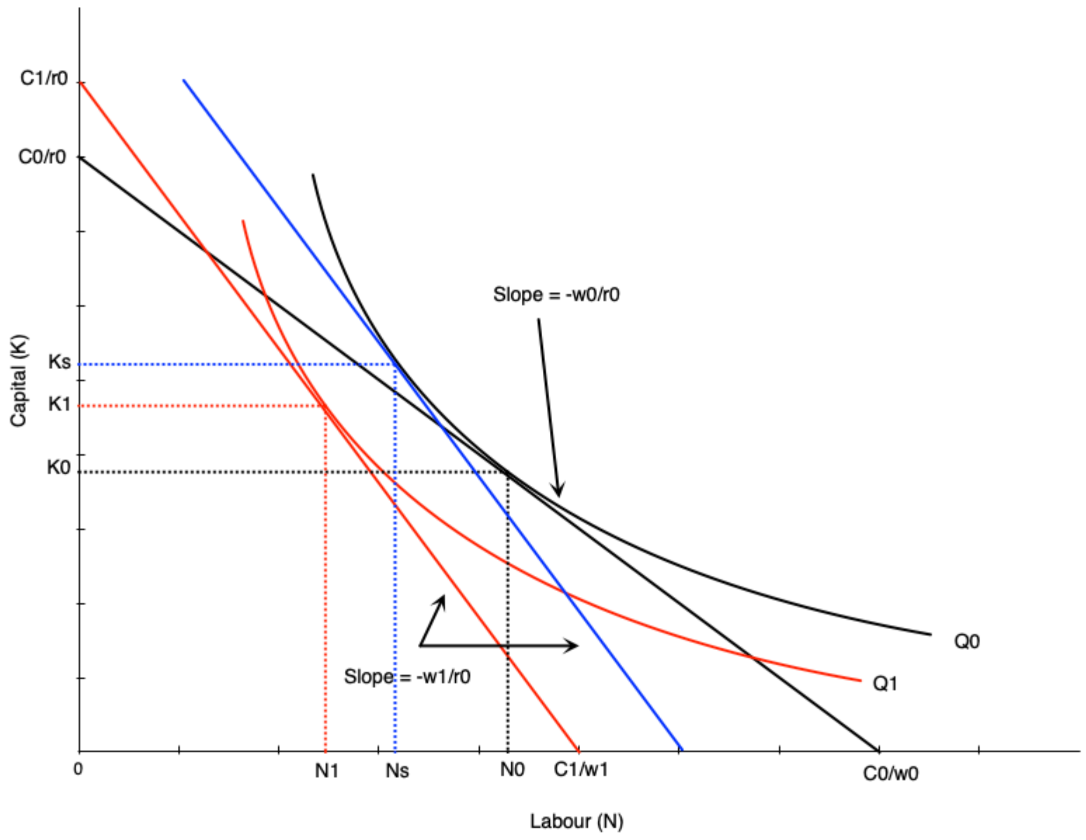
Deriving Labour Demand

- The labour demand curve summarizes the optimal choices of labour at different wage rates
- Graph in previous slide shows to two points on the labour demand curve
  - (w_0, N_0) and (w_1, N_1)
- To derive the full demand curve you can vary the wage rate to other levels
- Recall that the demand curve shows the amount of labour demanded at every wage *all else equal*
  - Other determinants of demand will shift the curve

Scale and Substitution Effects

- When the wage changes, two effects occur that influence labour demand
  - **Substitution effect:** change in input mix due to change in relative input prices
  - **Scale effect:** change in output due to change in cost of production
- The substitution effect makes firms want to substitute capital in place of labour
  - Leads to less labour demanded when wage rises
  - More capital demanded
- The scale effect comes from reduction in output, and leads to less labour demanded when wage rises
  - As production becomes more expensive, firms produce less
  - Use less labour and capital
- For labour demand, scale and substitution effect work in the same direction
  - But opposite directions for capital

Scale and Substitution Effects



- To find substitution effect, find cost minimizing bundle at new wage and old output
  - Blue line in the graph
  - Scale effect is $N_0 \to N_s$
- Scale effect is the change from optimum at new wage and old output to new wage and new output
  - Shift from blue to red line in graph
  - Scale effect is $N_s \to N_1$